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A data-driven method for optimizing the energy consumption of industrial robots

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ABSTRACT

With strong reliability and maneuverability, industrial robots play an essential role in the transformation and upgrading of the manufacturing industry, known as industry 4.0. Due to their mass application, it is essential to optimize the energy consumption of industrial robots considering the rising cost of energy resources and deteriorating eco-environment. State of the art solutions have mainly focused on hardware improvement and trajectory-planning-oriented software approaches, implementation of which is, nevertheless, restricted by the commercial feature of control algorithms of industrial robots. In order to explore the potential on energy saving, a data-driven approach for practical energy consumption optimization is proposed in this paper. Different from traditional energy consumption modelling where kinematic and dynamic behaviors of industrial robots are the major concern, this study use an artificial neural network to accurately reveal the quantitative relations between operating parameters and energy consumption, then genetic algorithm to optimize the adjustable parameters to minimize the consumed energy. Experiments are designed and performed on an Epson C4 6-DoF industrial robot. The results demonstrate the applicability and effectiveness of the proposed method in assisting the operating parameter selection and energy saving.

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1. Introduction

The past decades have seen a growing concern on promoting energy efficiency due to the rising cost of resources and the deterioration of the eco-environment. According to the statistics of the international energy agency, about 37% of energy supplied globally is consumed by industry (IEA, 2018b). Fortunately, many international organizations and governments, including Europe, the USA, China and so on, have invested considerable funds to improve the energy efficiency in fields such as manufacturing, chemical engineering and the like (Thomas and Rosenow, 2019; Wang et al., 2020). It is anticipated that there will be a twofold increase in the output per unit energy consumption by 2040 (IEA, 2018a). This will pose a great challenge to industrial robots (IR), which have enjoyed wide applications in modern manufacturing and will continue to play a contributing role in industry 4.0 and smart manufacturing for their reliability, controllability and applicability. Energy consumption by IR in workshop will dramatically increase considering its broad application and global sales growth in the near future (Acemoglu and Restrepo, 2017; Chiacchio et al., 2018; Paulus, 2014). Therefore, it is vital to optimize the energy consumption (EC) of IR and meanwhile improve its efficiency.

IR's energy saving optimization methods can be classified into three categories: Hardware, software and mixed solutions (Carabin et al., 2017). Hardware solutions involve optimal redesign, replacement or affiliation of components of an IR system. More efficient or lighter components substitute the original ones reducing the weight and inertia of arms whose movement, as a result, consumes less energy. Software solutions can be divided into two main subcategories: trajectory optimization and operation scheduling. Complex motion parameter-dependent models are constructed along with several predefined functions such as torque-rate, grid energy and effort supporting the modification of the path and motion profile in order to be more energy efficient. Operation scheduling generally considers the time scaling and sequence arrangement for a robot cell or robot production line. Mixed solutions are additional scenarios paying attention to both hardware and software modification of an IR system. The software method, due to its low cost and commonality, has been extensively studied and hence becomes the focus of this study.

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As mentioned above, software solution to energy consumption optimization for a single industrial robot mainly focus on trajectory optimization in which many objectives, including travelling time (He et al., 2016), energy-consumption (Field and Stepanenko, 1993), joint smoothness (Yang et al., 2008), etc. are taken into account to obtain the optimal motion profile of the IR. Generally, most of these studies calculate energy consumption through mathematical modelling, from simple models considering only the main dynamic part of industrial robot (Pellicciari et al., 2013) to later more and more complex models that incorporating rectifier (Xu et al., 2018), electronic devices (Yin et al., 2019) and plentiful other subtle devices. Glen and Yury (Field and Stepanenko, 1993) proposed an iterative dynamic programming method to design minimum energy consumption trajectory, where the power consumption is equal to the sum of power dissipated in resistive elements and delivered by motors. Another energy-efficient trajectory optimization model was proposed by Hansen et al. (2012), who considered not only friction losses but also the losses of servo drives and inverters. Meantime, the energy exchange between the robot joints via the coupled DC-bus was also calculated for the existence of recaptured electrical power. Similarly, Davis and Marcello (Meike et al., 2014) obtained the energy-optimal trajectory by means of time scaling, which explicitly considered the contribution of weight balancing mechanisms, electric drive dynamics, and several sources of energy loss in industrial robot system.

Recently, due to simulation capacity provided by commercial software, the simulation model driven software solutions are gradually favored by researchers. Some of these simulation tools are developed by IR manufacturers, such as Robot Studio® by ABB, KUKASim® by KUKA, etc., and others are commonly used mathematical or simulation tools including Matlab® and Delmia® with packages and toolboxes backing up the robot simulation. Abdullah and Bernard (Mohammed et al., 2014) addressed the problem of inverse kinematics and then dynamics through the analysis of forces and torque to get the feasible joint configuration. The lowest power consumption arrangement was chosen as the final one in which the power flow equals the product of torque and angular velocity. They compared the results of proposed model with a Robot Studio®-based one to show the superiority of their method. An IR simulation model that computes power losses in IR model, body mechanism, motor drive and control system under Catia® environment was proposed in (Paryanto et al., 2015). The measurement unit with 10 KHz of sampling frequency was applied to assist experimental investigation for the validation of simulation results. It was shown that there was an approximately 6% power deviation under predefined scenarios, which was mainly caused by the simplification of the simulation model.

Trajectory scheduling strategies and algorithms are typically inserted into the robot cabinet and protected as trade secrets by robot manufacturers. In other words, this kind of optimization process exists in the period of IR system design. Optimization of the motion profile is only accessible to the manufacturers of industrial robots and theoretically a matter of choice for the robot operator. Besides, typical optimization of EC of IR relies heavily on the modelling of IR system which depicts the relationship between parameters of each joint and final EC with kinematics and dynamics constraints. As the IR system becomes more complex, a growing number of components contribute to the EC of IR. There are always some parts of energy that cannot be modelled accurately as a result of simplification of the Permanent Magnet Synchronous Motors loss, friction loss and other related loss causing an apparent gap between optimal results and measured power flow from grids exits. This inaccuracy obviously affects trajectory strategies and energy optimization.

Therefore, this paper focuses on the software solution for a single industrial robot's optimization on EC from a totally distinct perspective, optimizing operating parameters, which has been proposed by a few researchers to assist the practically optimizing activity (Bowels et al., 2016; Rassõlkin et al., 2011; Riazi et al., 2017). The operating parameters, such as the maximum velocity and maximum acceleration, are usually determined by operators through robot teaching after frequent attempts or with the assistance of simulation models. Software solutions on the optimization of operating parameters rely either on simulation models or experimental measurements. The established simulation model can be found in (Gadaleta et al., 2019) in which the energy consumption of IR is computed using the module named Delmia Robotics® on the foundation of a Matlab®-implemented industrial robot model. The authors analyzed the motion parameters of the acceleration limit and velocity limit to trade off the energy consumption and motion time conceived for real-life industrial use, where the nonlinear relationship between the motion parameters and optimization objectives was proven. Along with the development of advanced sensoring technology, it is feasible to measure the electrical power flow between the main grids and IR cabinet which accurately reflects the overall energy that the industrial robot consumes. Experimental measurement can be noted in (Bowels et al., 2016), where operating parameters of a specific robot are analyzed with the variation of maximum velocity and acceleration and, revealing the higher velocities and acceleration require more power during the robot moving time. Another experimental measurement (Rassõlkin et al., 2011) was carried out to investigate the effect of distinct payload, velocity limit and workpiece position under a predefined motion profile of specific industrial robot, IRB 1600.

Though experimental measurement methods have offered sufficiently accurate power flow of IR system, the measured data are not studied comprehensively since only some simple rules are generalized in previous researches and they are not able to guide the determination of operating parameters. Owing to computer and data science, some data mining methods have been applied in the research area of energy consumption optimization or prediction of machine tools, petrochemical industries, even regional or continental activities, such as (Antanasijević et al., 2015; Geng et al., 2020; Yan and Li, 2013). Weighted grey relational analysis and response surface methodology were applied by (Yan and Li, 2013) to optimize the cutting parameters in the milling process in order to optimize energy consumption, production rate and cutting quality simultaneously. Antanasijević et al. (2015) established a general regression neural network model to predict energy consumption utilizing general country statistics as input, meanwhile analysis of energy-related greenhouse gas emissions. Pino-Mejías et al. (2017), taking Chile's tertiary buildings as a target, proposed an innovative utilization of artificial neural networks (ANNs) to forecast energy demand, energy consumption and CO_2 emissions. ANNs, inspired by the signal processing and transformation of neurons in brains (Taki et al., 2018), are regarded as an analytical method of simulating system performance. They proved to be capable of assisting the complex relationship between parameters and objectives, and thus are frequently used in the estimation of EC (Elsheikh et al., 2019). It is also proved that the multi-layer neural networks are in a position to approximate any linear or non-linear function with a specific accuracy (Sun and Gao, 2019).

Therefore, to identify the quantitative features between operating parameters and energy consumption, an energy model of IR based on ANNs is proposed in this paper for precise and practical analyses of IR energy consumption without specifying the mathematical relationship between the input and output variables. Since the structure and various parameters of the neural network model make it difficult to obtain the optimal EC value through mathematical calculation such as partial derivative. The genetic algorithm

(GA), a kind of stochastic search methods inspired by the evolutionary strategies in biological evolution, has been a choice of method by many researchers and been extensively used either as a stand-alone algorithm or as a primary algorithm in hybrid approaches (Defersha and Rooyani, 2020). Therefore, GA is selected to look for the parameter values for its capability of global search and effectiveness in NP-hard problem solving in our study (Yan et al., 2016). The energy prediction model is established as the fitness function of GA to search for the optimized parameter combination in variable spaces. With the advanced sensoring and interconnecting technology, data are easily obtained in the era of Industry 4.0 allowing the data application in the construction of intelligent factory and execution of intelligent management. The proposed data-driven method is capable of exploring the potential of operation data of IR system into energy consumption optimization, and hence in the sustainability of manufacturing.

This method is proposed for the guidance of the environmental-friendly operation of industrial robots, and for the quick modelling of the EC of IR system without any need for deep understanding of the robot's kinematic and dynamic behaviors. The rest of this paper is organized as follows: Section 2 describes the framework of the data-driven optimization method and the detailed algorithms; Section 3 provides a case study to demonstrate the validity and effectiveness of the proposed approach, and Section 4 makes the concluding remarks and discusses about future directions of improvement.

2. Data-driven optimization method of EC

The typical industrial robot system consists of three parts, manipulator, drive and control system. Under the guidance of the control system, the drive system forces the manipulator to execute some motions accurately to assist production activities. The manipulator usually contains mechanical components, gearings, permanent-magnet synchronous motor, etc. The drive system comprises rectifiers, inverters and so on. The control system includes mainboards, fun, processors and related electronic devices. All these parts contribute to the energy consumption of an IR. Consequently, EC of industrial robot system can be obtained by monitoring the power demand between the cabinet and main grids. Fig. 1 depicts an industrial robot's power profile under three cycles of certain motion commands during a period when the end-effector of robot moves straight from the same starting point to the same destination and then returns. The maximum value of velocity and acceleration in the control programs of cycles I, II, III are shown in Table 1.

Table 1
Predefined operating parameters of each cycle.

Cycle	Maximum velocity (%)	Maximum acceleration (%)
I	5	50
II	10	50
III	15	50

As shown in Fig. 1, the profile suggests three different states, stand-by, servo-start and active working. When the IR system powers up, it enters the stand-by stage. In this period, only control system consumes energy. Before motion, the control system sends commands to initiate servos. In the servo-start stage, power flows of control system and manipulator are approximate constants. As the sketch shows, in the working stage, the power flow rises and the motion time drops in each cycle since the operating parameter, the maximum velocity noted as V_{max}, increases gradually indicating that the defined velocity threshold in motion command indeed affects the EC. Besides the V_{max}, fundamental operating parameters in control programs of an IR also contains the maximum accelerations A_{max}, coordination of end-effector *Posi* and motion modes *mod*, including point to point movement (PTP) and continuous points movement, where PTP mode is applied most commonly to allow the end-effector to move in the fastest trajectory without constraints on the trajectory shape and continuous point mode, on the contrary, the shape of trajectory to be linear or cycle. These parameters also impact the energy consumption. Their non-linear relationship with energy consumption of a free load IR system is explained through data analysis in this paper.

The optimization method of this study is based on experimental investigations and data measurement to ensure accurate modelling of EC of IR system. The steps of operating parameter optimization are shown in Fig. 2. Firstly, operating parameters that affect the EC have to be determined, which has already been introduced.

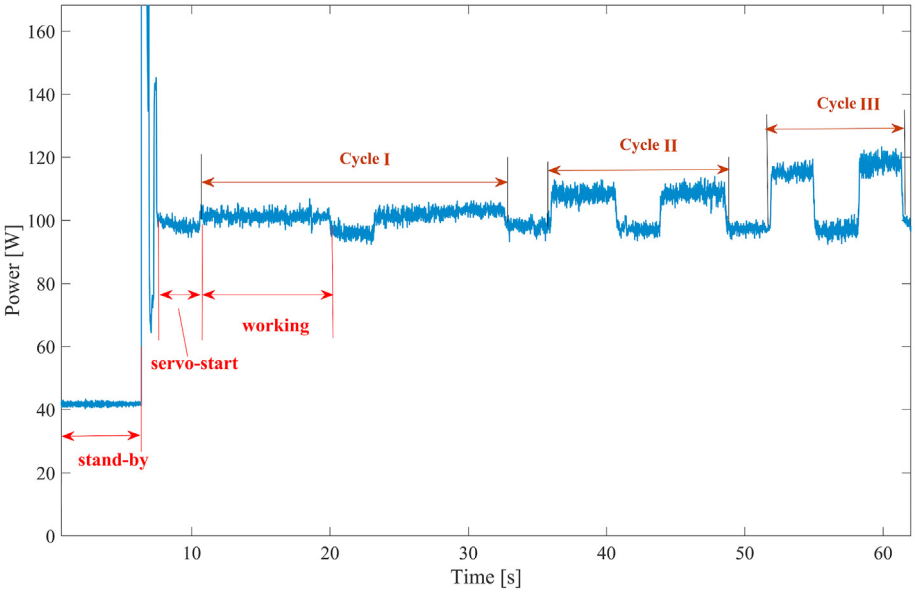


Fig. 1. Power profile of a motion process.

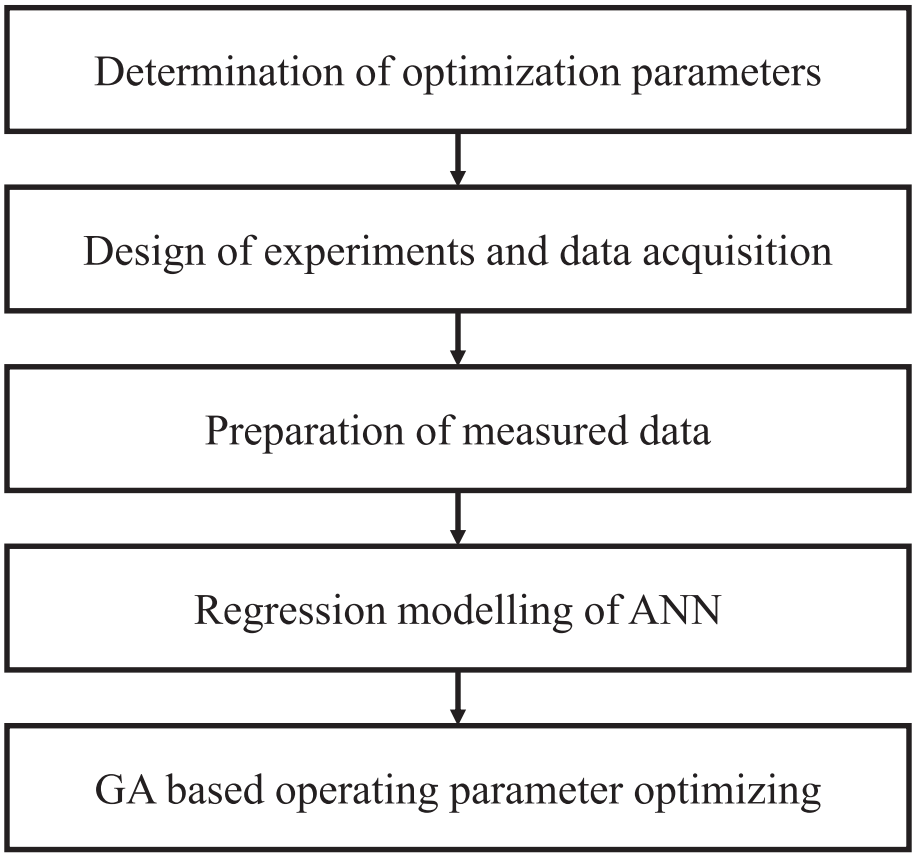


Fig. 2. Flowchart of operating parameter optimizing.

Secondly, the experiments are designed according to the variables and testbed is set up to measure the power flow of the IR system. Next, running logs of IR are analyzed from which time stamps are extracted to ensure the starting and finish time of the manipulator motion on the purpose of calculating the EC through the integral with power profile measured in the previous step. Besides the computation of consumed energy, the samples are also normalized in the light of the preprocess of variables in order to facilitate the modelling process and increase the precision of EC prediction.

The ANN regression theory, due to its capability of complex relationship structing, is then applied to establish the relationship between operating parameters and corresponding consumed energy. It is reported that the neural network transfers signal through the neuron, which, first receives signals from other connected neurons with weight, then compares with the bias, and finally outputs values to next neurons through an activation function. The primary task of an ANN is to obtain the weights and biases in a way that minimizes the error between the output value and the actual one with the same input. The back propagation (BP) algorithms, with a multi-layer network structure, are validated to be a highly effective way for model training by (Liang et al., 2020) who utilize the back propagation of output error to adjust the weights and biases automatically along the steepest descent path for the purpose of minimizing the error to the desired value.

Given the structure of operating parameters and complexity of model training, the BP neural network is used in our research (Fig. 3), which is a three-layer network containing an input layer, a hidden layer and an output layer with only one neuron in it to represent the prediction value of EC under corresponding operating parameters. Weight among these layers is marked as w_{ih} and v_h respectively, where i is the neuron number of input layer and h is the number of neurons in hidden layer. Meanwhile, biases of neurons in the hidden layer and output layer are depicted as α_h and β respectively. Then, the relationship between the output of the neural network $\widehat{W}$ with a vector of inputs $OP = (V_{max}, A_{max}, Posi, mod)$ becomes:

$$\widehat{W} = g_2\left(\beta + \sum_h v_h \cdot g_1\left(\alpha_h + \sum_i w_{ih} \cdot OP\right)\right) \tag{1}$$

where g_1 and g_2 are the activation functions for hidden and output layer respectively. Consequently, for a given data set $D = \{(OP_k, W_k)\}_{k=1}^m$, the mean square error adopted to evaluate the accuracy of the BP model is calculated as:

$$MSE = \sum_{k=1}^{m} (W_k - \widehat{W_k})^2 \Big/ m \tag{2}$$

where W_k is the measured value of EC for kth vector of OP and m is the number of samples that the set contains.

Algorithm 1 describes how the parameters of BP neural network are computed and updated and the trained neural network along with its parameters forms the EC model of IR system. Detailed training parameters of the network, including the learning rate, activation function and so on will be discussed later.

The last step of the proposed method, as described in Fig. 2, is to find optimal operating parameters to reduce the EC based on the model trained in the previous step. Thereby, the neural network model is recognized as objective function and the operating

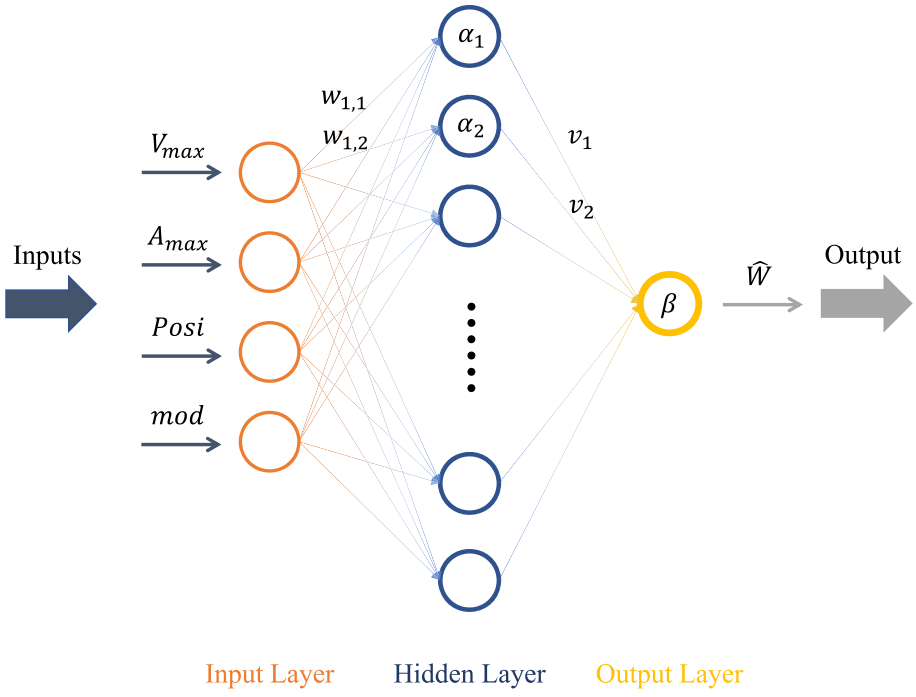


Fig. 3. BP network topology.

parameters, in turn, are supposed to be the results of the optimization solution for obtaining the combination of input parameters to minimize EC. As mentioned above, some of operating parameters are actually tasks determined and cannot be set randomly, while the remaining parameters are adjustable. Thereby, chromosomes of GA are designed according to those adjustable operating parameters. Algorithm 1 shows the detailed process of GA-based parameter optimization.

Algorithm 1. .

Input: Data Set $D = \{(OP_k, W_k)\}_{k=1}^{m}$;
Divide the data set into training D_{train}, validation D_{val} and test D_{test} sets;
Initialize weights and biases of the BP neural network randomly within (0,1);
while the terminal conditions are not reached, do
  for all (OP_k, W_k) in D_{train}
    Compute the output $\widehat{W_k}$ of BP neural network to get MSE_{train};
    Calculate the gradient of output layer and then the gradient of hidden layer;
    Update weights and biases of the BP neural network;
  end for
  Generate the MSE_{val} and MSE_{test};
end while
Decide the adjustable operating parameters and the cost function;
Initialize population randomly within the range of operating parameters;
while the stop criteria are not reached, do
  Calculate the fitness values of individuals in current generation;
  Update individuals through elite selection, crossover and mutation.;
  Check the score of population based on the established cost function;
end while
Output: the best fitted individuals in the last population.

3. Optimization method implementation

3.1. Experimental details

To validate the proposed methods, experiments were conducted on a 6-DoF industrial robot system shown in Fig. 4 (Epson C4 industrial robot with a maximum load of 4 kg and motion range of 900 mm). Fig. 5 shows the experimental setup where the overall power demand of IR system was measured using an oscillograph recorder (YOKOGAWA DL350 with a maximum sampling frequency of 10 MHz) and probes were installed between the main grids and the IR cabinet. For the single-phase power prerequisite of IR system, one voltage clamp and one current clamp are required in the testbed for the investigation. The voltage clamp is able to measure up to 1000 V with respect to earth in a maximum of 60 MHz frequency and the current clamps up to 50 A with the sensitivity of 10 mV/A. A high sampling frequency is required to record subtle changes of power flow concerning the frequency of grid and impulsive flowing in electric devices. Measured data in our recorder were stored in the secure digital memory card in the form of binary, mat and so on or were transferred to PC based on the Modbus protocol. Weighing the pressure of data transfer and measurement precision, sample frequency of 100 Khz was ensured to acquire the power flow in experiments.

Experiments were carried out to investigate the influence of operating parameters on EC by varying three parameters V_{max}, A_{max}

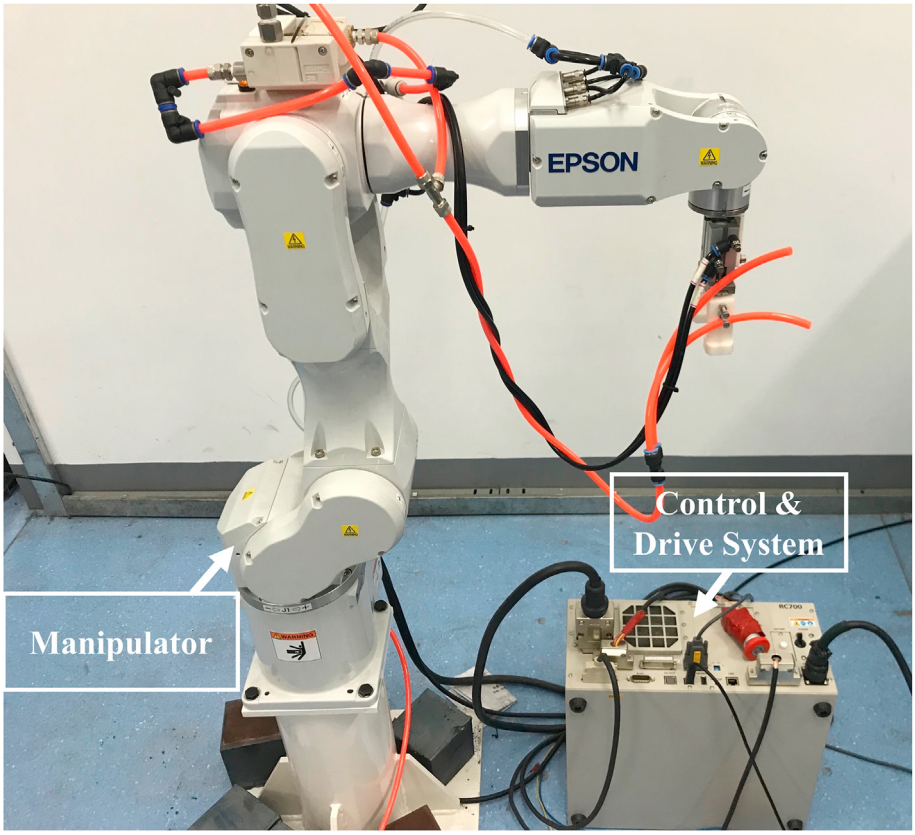


Fig. 4. Epson C4 6-DOF industrial robot.

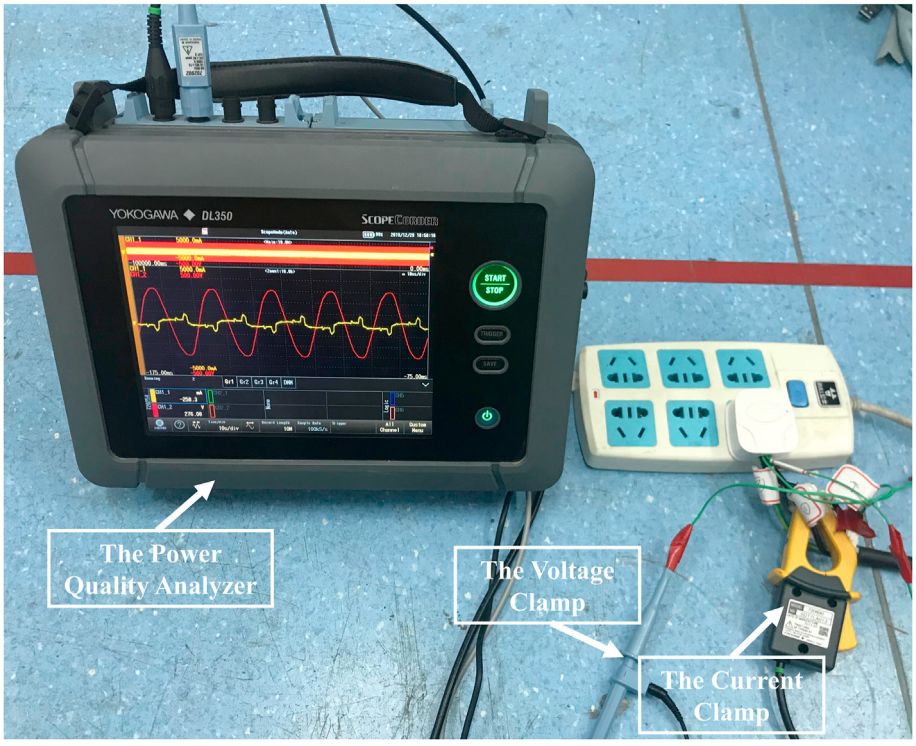


Fig. 5. The oscillograph recorder and its installation.

Table 2
Experimental design and ranges of operating parameters.

Case No.	V_{max} (%)			A_{max} (%)			*Posi*	Distance (mm)
	Minimum value	Maximum value	Step	Minimum value	Maximum value	Step		
1	10	90	10	10	90	10	P_1—P_3	300
2							P_1—P_5	600
3							P_1—P_2	900
4	10	90	10	10	90	10	P_6—P_4	450
5							P_7—P_2	
6							P_7—P_4	
7	10	90	10	10	90	10	P_2—P_8	779.42
8							P_9—P_1	
9							P_2—P_6	

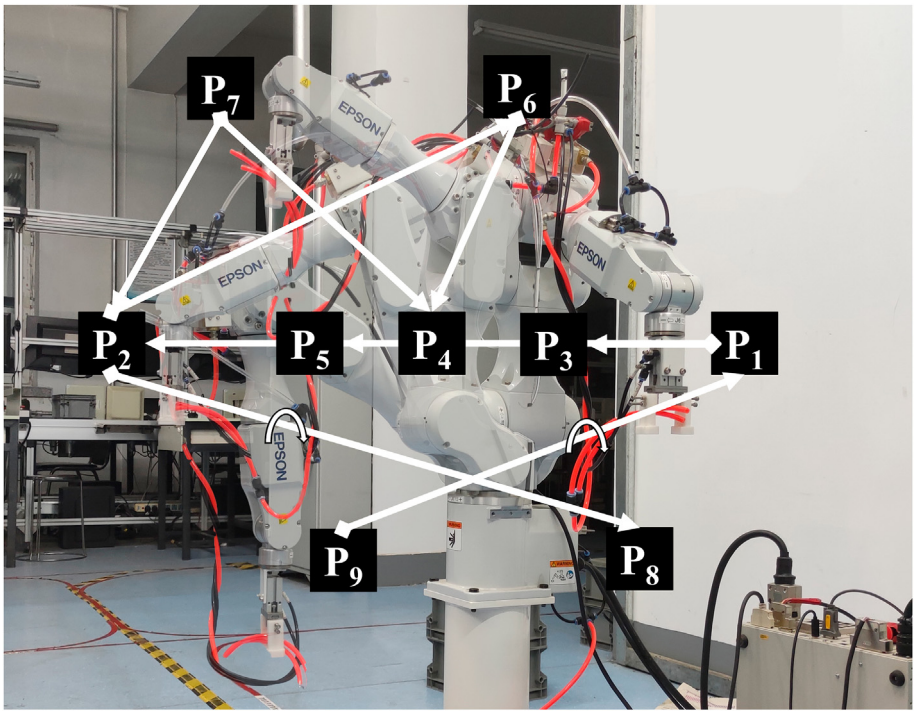


Fig. 6. Absolute coordinates of nine points for 9 cases.

Table 3
Absolute coordinates of nine points.

Posi	X(mm)	Y(mm)	Z(mm)	U(°)
P_1	- 500	0	+450	- 180
P_2	+220	+540	+450	- 300
P_3	- 260	+180	+450	- 220
P_4	- 140	+270	+450	- 235
P_5	- 20	+360	+450	- 260
P_6	−320	+135	+840	- 200
P_7	+40	+405	+840	- 280
P_8	−320	+135	+60	- 200
P_9	+40	+405	+60	- 280

and *Posi* in *mod* of PTP, the most commonly used one in practical circumstances, to enforce the end-effector of the selected robot. As Table 2 shows, V_{max} and A_{max} varied from 10% to 90% of their corresponding maximum values by a step of 10% and it should be noted that V_{max} and A_{max} are standardized in percentage under PTP mode. In terms of *Posi*, nine different cases selected in a hexagon were analyzed to explore the rules between the distance or direction of movement and the final EC, as shown in Fig. 6. The varying velocity and acceleration limitations produce 81 (9×9) samples for each case. Absolute coordinates of nine points in cartesian space are presented in Table 3 in form of (X, Y, Z, U) which were also directly used in assignment statements and motion commands. The first three cases, with the same starting point P_1 and moved in the same direction, were designed to illustrate the impact of moving distance on EC of an IR. Cases 4, 5 and 6 were performed to explore the influence of moving direction on the EC and the last three cases to determine the effect of direction and position using the complex actions of the manipulator (Fig. 6).

After the preprocessing of experimental measurements of all cases, EC is plotted separately taking velocity and acceleration limitations as coordinate axis in 3D maps (Fig. 7). The color bar in the right side represents the EC value with the range of [60,550] (expressed in J) and each map consists of 81 measured samples. The maps demonstrate an apparent surge in the overall EC as the movement distance increases. In each case, the downward trend of EC exists where the V_{max} dominates and subtle changes occur when the A_{max} and V_{max} increase continuously. Therefore, it is essential to establish an EC model to identify the optimal parameters to make EC minimized.

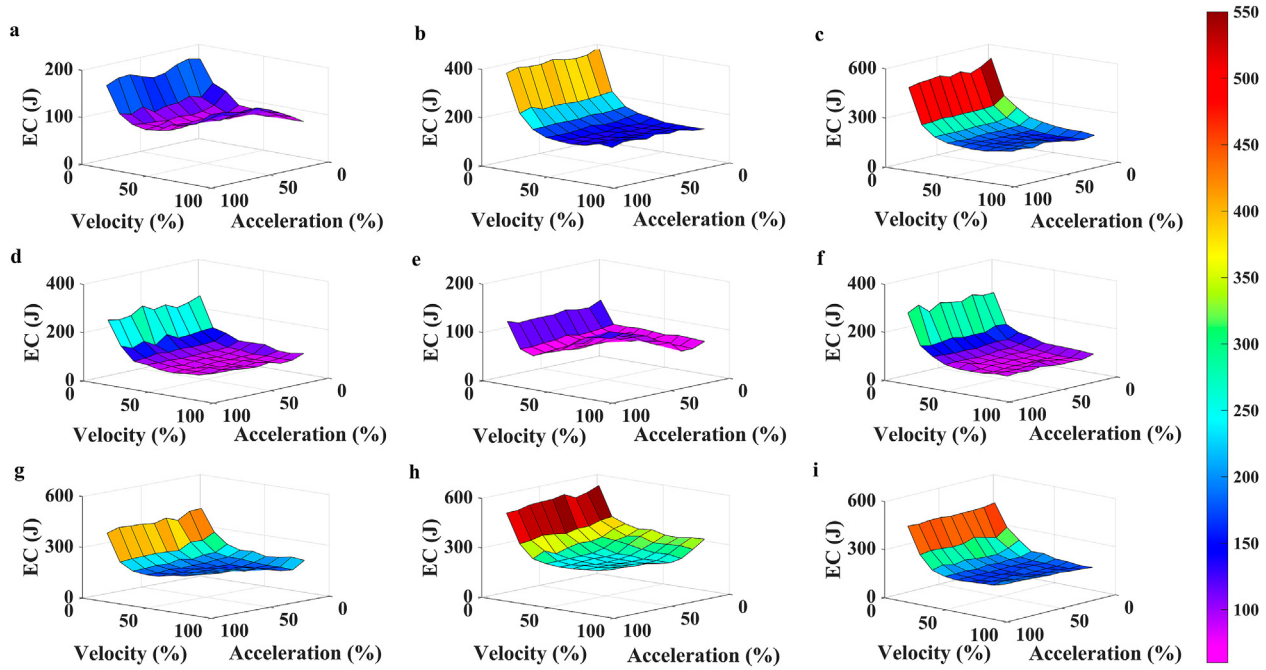


Fig. 7. 3D color maps of measured EC for 9 cases from (a) P_1–P_3 to (i) P_2–P_6. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 4
Parameters of BP neural network.

No.	Parameters	Value
1	Number of input layer neurons	10
2	Number of hidden layer neurons	16
3	Number of output layer neurons	1
4	Learning rate	$1*10^{-3}$
5	Training algorithm	Levenberg-Marquardt
6	Maximum epoch	200
7	Training convergence	$1*10^{-6}$
8	Transfer function	Sigmod
9	Validation Check	20

3.2. Prediction model of EC based on BP neural network

To describe the complex mechanisms and linkages between the operating parameters and EC, the BP neural network was applied to establish an accurate model. The input of the model was the vector of normalized operating parameter values in Table 2 and the output was the normalized EC under a certain scenario, which determines the number of neurons in the input and output layers to be 10 and 1 respectively.

A total of 729 (81×9) samples were designed and measured in our experiments and they were grouped into the training set, validation set and test set in the way of stratified sampling with 441, 144 and 144 samples respectively to promote the comprehensive learning and enhance the generalization ability of the network. Other parameters of the network are listed in Table 4. They were selected either through empirical formula or controlled experiments and detailed explanations will not be given here since parameter selection of BP is not focus of this research. The algorithm was coded in Matlab® software and an Intel® Core™ i7-8700 CPU @ 3.2 GHz computer with 8 GB RAM was used.

Fig. 8 represents the matching degree of the predicted EC with the measured one for each sample sets where the closer of the value of correlation coefficient (Noted as R) is to 1, the better the regression model is. The prediction results of the selected 144 test samples are demonstrated in Fig. 9 along with the measured data for comparison and the mean absolute value of the relative error of the test dataset is 4.59% indicating that the trained model has an excellent compatibility to the experimental data and is able to predict the EC with higher accuracy than other EC prediction method in (Paryanto et al., 2015) where the value of deviation is approximately 6%. The well-trained BP neural network finally ensures weights and biases under the designed network structure, which was further utilized to establish the non-linear cost function of EC optimization.

3.3. Optimization of EC based on GA

After establishing the complicated relationship between the operating parameters and EC through BP neural network along with its final parameters, GA was employed in this research to find the optimal solutions for its capability of global search and NP-hard problem-solving. Robot motion is commonly determined by the production requirements to facilitate logistics, processing, transportation and other activities for collaboration with or replacement of humans. In logistics and palletizing circumstances, starting and ending points of IR are firstly defined to accomplish the delivery or arrangement of product and the remaining operating parameters are fairly liberally set. Thus, the present study selects the velocity and acceleration limitation as adjustable parameters to verify the proposed method.

The chromosome was encoded in binary with fourteen genes depicting the varying velocity and acceleration limitation within the range of [10,90] in decimal integer concerning the round number requirement of motion program. An example of chromosome is shown in the vector in Fig. 10. Extra parameters of GA, proven to be suitable for our application, are listed in Table 5 and also. The GA also ran in the above environment.

When the stopping criteria of GA were reached, the best individuals in the last generation were obtained as final schemes and

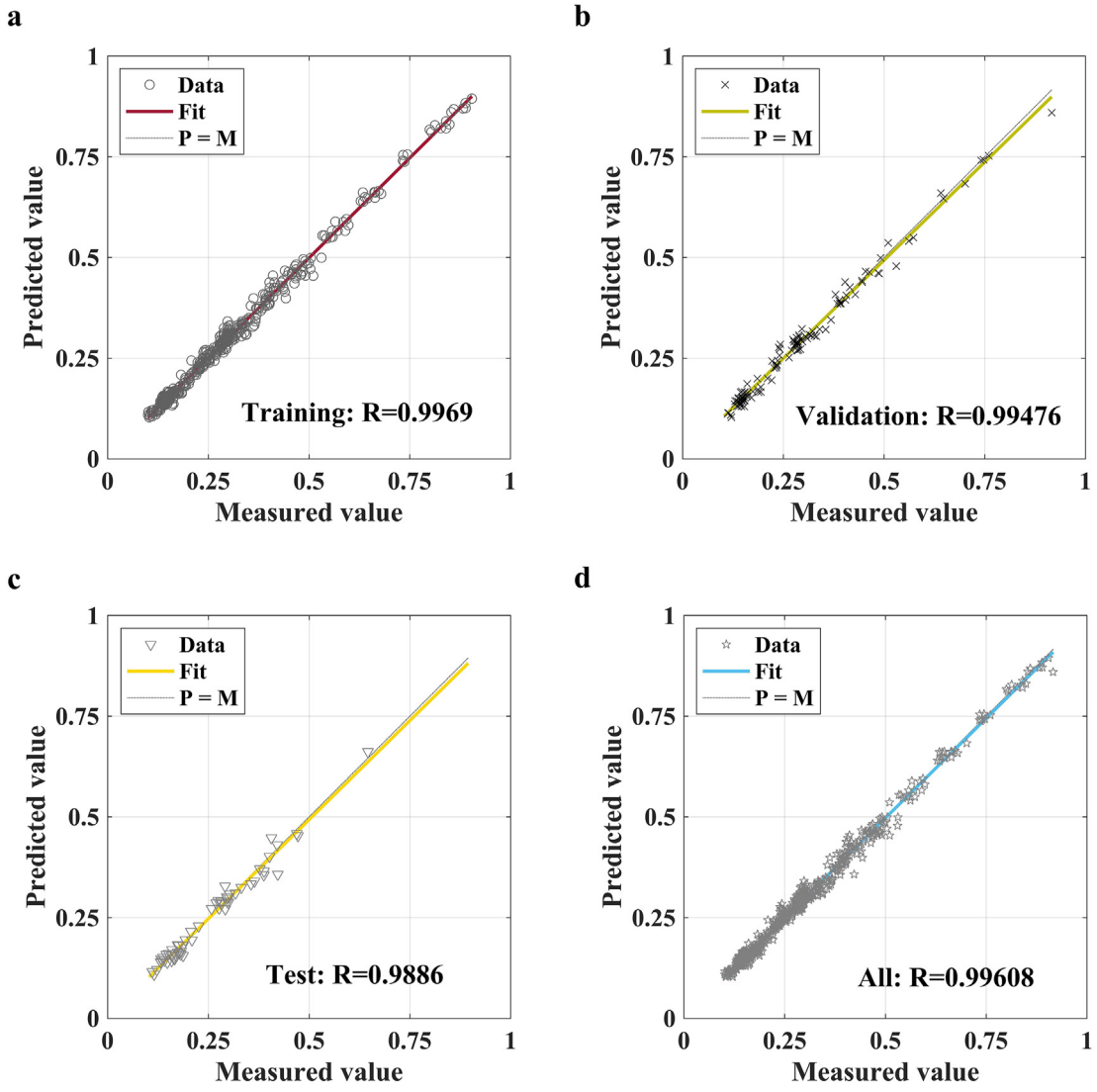


Fig. 8. Predicted vs. measured values of normalized EC and corresponding cross-correlations for: (a) training set; (b) test set; (c) validation set; (d) all samples.

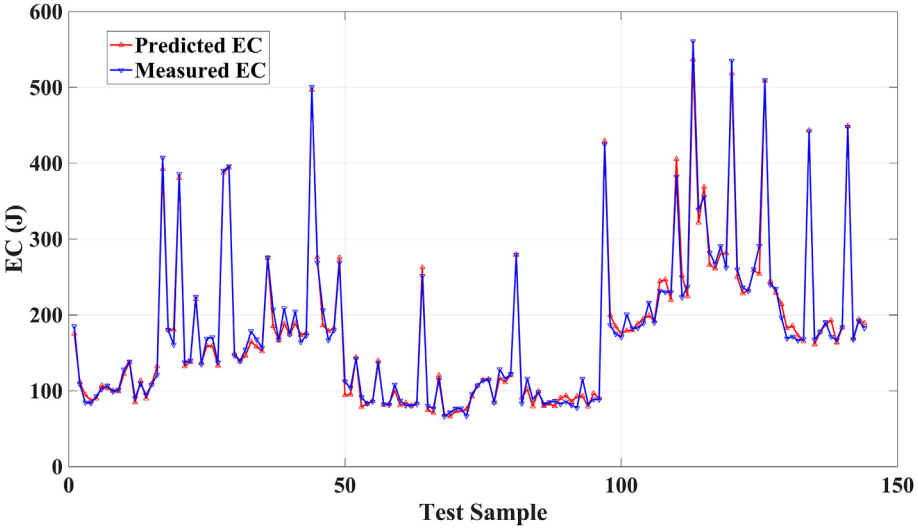


Fig. 9. Predicted vs. measured values of EC of test samples.

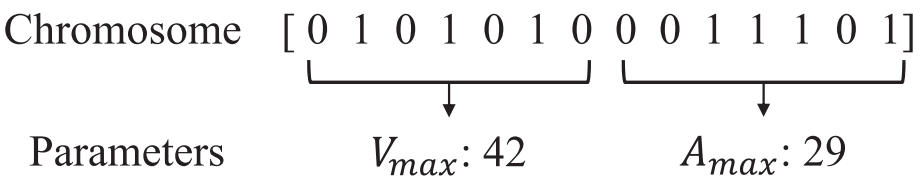


Fig. 10. An encoding example.

Table 5
Main parameters of GA.

No.	Parameters	Value
1	Population size	50
2	Elite selection	5
3	Maximum generation	200
4	Crossover rate	0.6
5	Mutation rate	0.1
6	Function tolerance	$1*10^{-6}$
7	Stall generation	50

the optimizing results of all scenarios were listed (Table 6) and marked on color maps as white spheres (Fig. 12). Taking the fifth case P_7—P_2 in Table 2 as an example. Profiles that illustrate fitness values of the best individual in each generation and their means are described in Fig. 11, representing the parameter optimizing process. The optimized results were verified by coding the parameters into motion command. The comparison of predicted-measured values is shown in Table 6 with a mean relative error of 4.56%, which indicates that the proposed method is capable of adjusting operating parameters to minimize the overall EC with high accuracy.

Fig. 12 shows the predicted values of EC with the optimal combination of operating parameters for each case. Compared with Fig. 11, it indicates the possibility to cut down the EC by 52.32%, 66.30%, 70.76%, 69.32%, 49.57%, 71.08%, 61.34%, 59.21% and 62.95% respectively, where the measured EC varies from 179.59 J, 406.90 J, 542.71 J, 269.06 J, 125.55 J, 280.98 J, 424.78 J, 561.01 J, 456.22 J (IR operating at 10% of velocity and acceleration limitation) to values of 85.62 J, 137.12 J, 158.69 J, 82.56 J, 63.32 J, 81.25 J, 164.21 J, 228.83 J, 169.03 (IR operating at optimized velocity and acceleration) (Fig. 13). Thus, the effectiveness of the proposed method is verified by these results.

3.4. Results and discussion

This section discusses the quantitative level of the EC and their tendency under different cases. It is illustrated that the maximum EC happens at the moment when both the value of velocity and that of acceleration are minimal, for the reason that operating time is the main influencing factor for the EC when the time is comparatively long and power flow relatively small, as is shown in Fig. 12. Profiles in Fig. 14 simulating the motion of end-effector of IR depicts values of velocity under time domain and shows the basic mathematical relationships among velocity, time, acceleration (derivation of velocity versus time), and distance (integral of velocity versus time). A comparison between trajectories 1 and 2 shows that the

Table 6
Comparison of measured and predicted values on optimized operating parameters for all cases.

Case No.	V_{max} (%)	A_{max} (%)	Predicted EC (J)	Measured EC (J)	Relative error (%)
1	32	68	81.81	85.62	4.45%
2	53	71	131.56	137.12	4.05%
3	90	18	150.57	158.69	5.12%
4	53	24	76.82	82.56	6.95%
5	26	78	60.04	63.32	5.18%
6	43	88	77.17	81.25	5.02%
7	43	35	161.51	164.21	1.64%
8	45	90	217.51	228.83	3.92%
9	51	47	161.03	169.03	4.73%

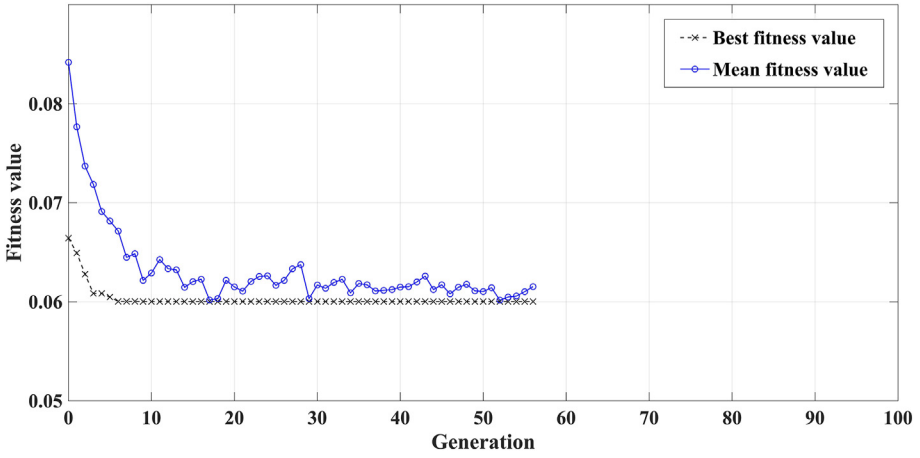


Fig. 11. The convergence curves of GA while optimizing the parameters of the fifth case P_7—P_2.

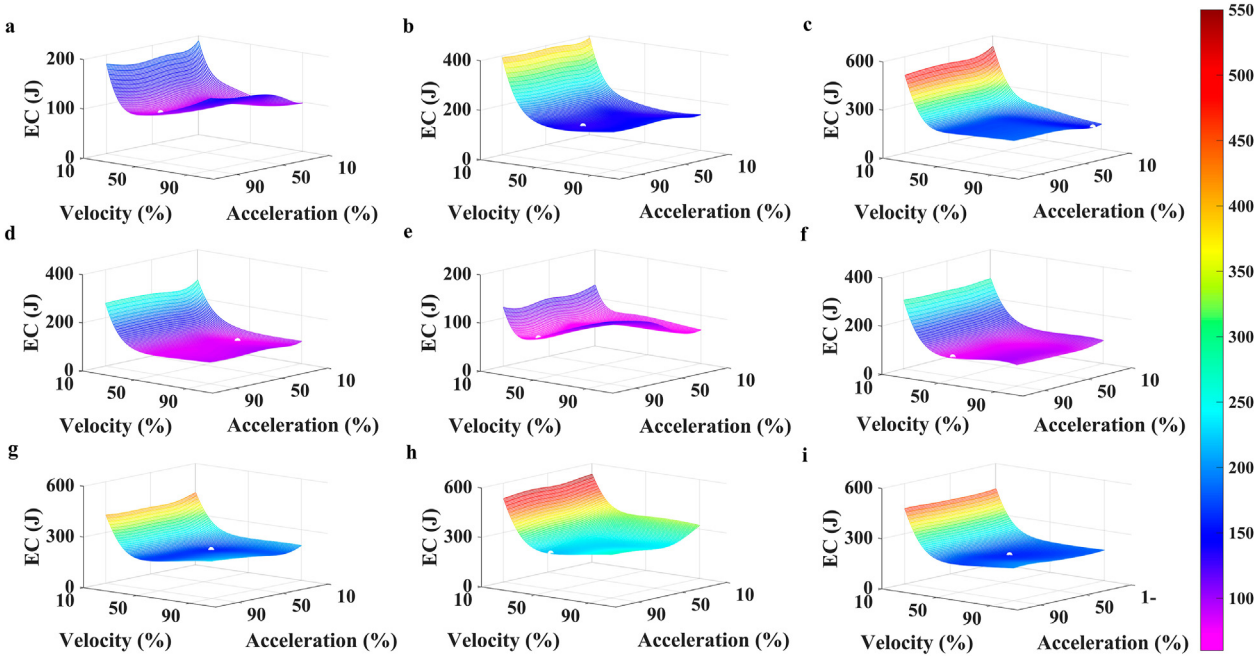


Fig. 12. 3D color maps of predicted EC with optimal solution marked for 9 cases from (a) P_1–P_3 to (i) P_2–P_6. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

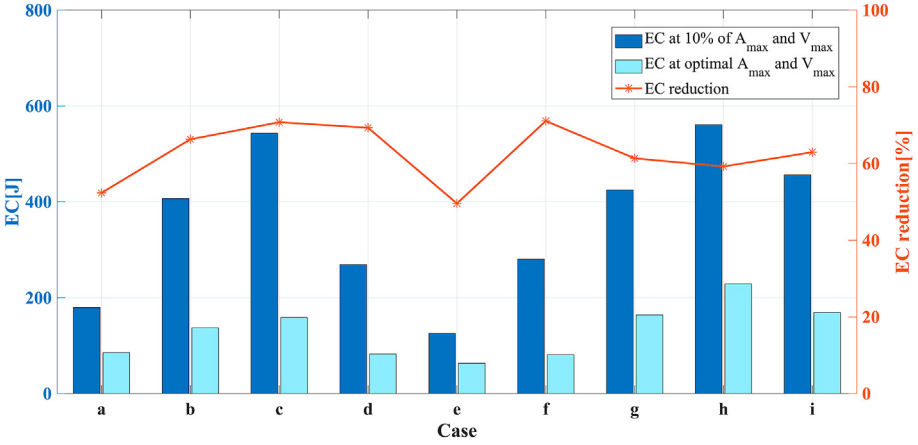


Fig. 13. The comparison of EC before and after the optimization.

time increases by 62.5% in order to reach the same destination when the velocity limitation reduces to half. When acceleration limitation decreases, time gains the increment are shown by trajectories 1 and 3. Also, trajectory 4 implies the possible situation where the velocity limitation is not reached, causing the running time to double. Therefore, the level of EC is relatively high since the long running time with low power demand and moderately enhanced velocity and acceleration limitation interplay to reduce the EC of IR.

However, the continuous rising velocity and acceleration limitation leads to a surge of power demand, which furthers the EC since the time fluctuation is too narrow to have a dramatic impact at this moment as indicated by trajectories 1, 5 and 6. Then, the optimal operating parameters appear to trade off the power flow and running time, allowing the EC to be minimized, as shown in Table 6 and Fig. 12. Cases 1, 2 and 3 illustrate the impact of moving distance and it is obviously shown that with the increase of distance, the EC of the IR also surges. From cases 4, 5 and 6, we can see that the starting point and the moving direction also influence the EC since both the shape of the profiles and optimal parameters differ. As for cases 7, 8 and 9, besides the moving direction and starting point, the multi-joint movement is also proved to enhance the EC level. Therefore, the proposed method is able to take these factors into account by using velocity limitation, acceleration limitation and the coordination of end-effector as the input of the prediction model.

In fact, this paper addresses the issue of the EC optimization of IR system aiming at the guidance of operating paraments selection in the actual operation. The regular pattern revealed in our research for the IR EC is distinct from others (for example (Gadaleta et al., 2019; Paryanto et al., 2015)) due to the diversity of IR's structure, components, etc., which also confirms the necessity for EC optimization in a practical way and the effectiveness of our proposed approach.

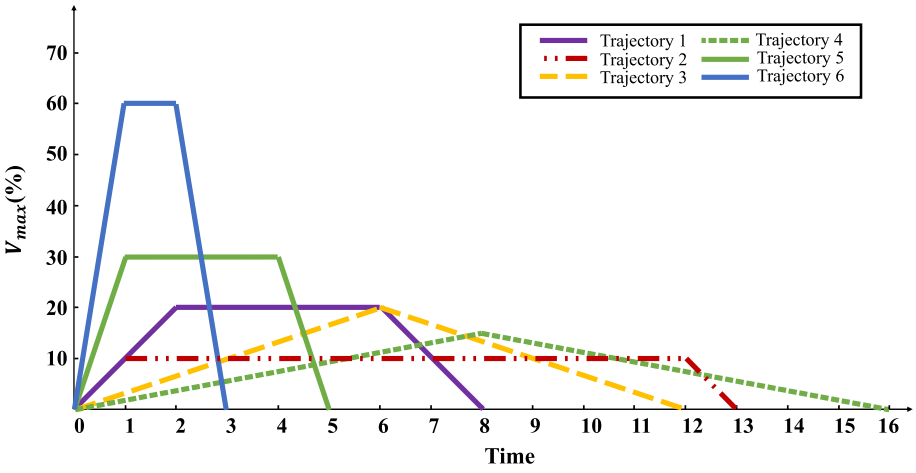


Fig. 14. Velocity of end-effector with different limitations on operating parameters.

4. Conclusions and future research

In this study, a data-driven optimization method for energy consumption of IR system is proposed to guide the determination of the combined operating parameters. The relationship between the operating parameters and EC has been revealed using the BP neural network's structure and variables, where mass measured data are preprocessed and grouped for the training, validating and testing processes to obtain an accurate prediction model. Then, it is regarded as the target function and GA is applied to efficiently identify the optimal adjustable parameters in order to minimize the energy consumption. Experiments on Epson C4 with the assistance of the oscillograph are performed. The decreased EC together with the accurate model firmly validates the effectiveness and efficiency of the proposed solution. Compared with the EC model based on kinematic and dynamic behaviors of industrial robots, the ANNs and GA based optimization method utilize machine learning to replace the complex model, which not only enhances the prediction accuracy but also enlarges the application scope of EC saving. In particular, it is noteworthy that the data-driven method for the energy consumption of industrial robots is presented for the first time. By addressing the easily ignored operation data of IR system and enhances the value of information in a comprehensive way, this method is able to guide the operation of industrial robots and ensure their environment friendly characteristics.

Future research will extend the current scope of operating parameters in the BP neural network (e.g. the load on end-effector, other complex motion mode) for a comprehensive energy consumption model of IR. In view of the wild application of the proposed approach, it is also urgent to incorporate the optimizing method into cloud services in the era of computing and communication technologies to benefit both the industrial robots' manufacturers and users.

CRediT authorship contribution statement

Mingyang Zhang: Conceptualization, Methodology, Formal analysis, Data analysis, Writing - original draft, Investigation, Software. **Jihong Yan:** Validation, Investigation, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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